AUSTIN ANIMAL CENTER: BAYESIAN CLASSIFICATION

Mixed Naive Bayes Classifier (Categorical & Gaussian Features)

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1. Overview

Animal shelters face critical operational challenges in anticipating the outcomes of admitted animals. Predicting outcomes at intake supports targeted medical intervention, behavioral rehabilitation, and capacity allocation. In this work, we implement a from-scratch mixed Bayesian classifier combining discrete categorical demographic attributes with continuous duration metrics to predict shelter outcomes across approximately 80,000 historical intake records from the Austin Animal Center (AAC). Adhering to strict academic constraints, all preprocessing, probability estimation, and evaluation are implemented purely in numpy and math without third-party ML or data frameworks.

2. Preprocessing

Strict rule-based transformations were enforced without data fabrication:

- **Four-Class Target Collapse:** Mapped the outcome field into {Adoption: 42.17%, Transfer: 29.87%, Return_to_Owner: 18.57%, Other: 9.40%}.
- **Derived Age in Days (age_days):** Parsed raw intake age strings using standard unit multipliers (years=365, months=30, weeks=7, days=1; unknown/invalid=0).
- **Derived Stay Duration (time_in_shelter_days):** Computed ceil(OutcomeDate - IntakeDate) via a calendar-accurate astronomical day-counter implemented strictly with integer and string operations.
- **Categorical Encodings:** AnimalType (Dog=1, Cat=0, other=0); SexUponIntake (Neutered/Spayed=2, Intact=1, other=0); Color (uppercase first token); and Breed (top-20 most frequent from TRAIN only, mapping others to 'Other').
- **Stratified Splitting & Scaling:** Stratified 80/20 train/test split (Train=63,737, Test=15,935) with fixed seed (242013 % 97 + 7 = 102). Numeric features were z-scored using mean and variance derived strictly from TRAIN with variance clamped at ≥ 1e-6.

3. Model Architecture

The classifier models joint class-conditional log-posteriors assuming naive conditional feature independence:

$$\ln P(c \mid x_{cat}, x_{num}) = \ln P(c) + \sum \ln P_{cat}(x_f \mid c) + \sum \ln N(x_j; \mu_{c,j}, \sigma^2_{c,j})$$

- **Laplace Smoothing:** Applied student-derived constant K = 0.5 + ((sum of digits of 242013) % 5)/10 = **0.70** to all categorical feature count tables.
- **Gaussian Continuous Likelihood:** Estimated class-specific mean and clamped variance for age and duration, computing standard normal log-density.
- **Numerical Normalization:** Stable Log-Sum-Exp normalization prevented underflow across all posterior calculations.

4. Empirical Results

Evaluation on the 15,935 held-out test records achieved an **Overall Accuracy of 55.23%** and a **Macro-F1 of 0.5629**.

Class Name	Precision	Recall	F1-Score	Test Support
Adoption	0.6124	0.5156	0.5598	6,719
Transfer	0.5037	0.5821	0.5401	4,760

Return to Owner	0.4671	0.6288	0.5360	2,958
Other	0.8869	0.4713	0.6155	1,498
Macro Average / Total	**0.6175**	**0.5495**	**0.5629**	**15,935**

Actual \ Pred	Pred Adoption	Pred Transfer	Pred Return	Pred Other
Actual Adoption	3,464	2,006	1,226	23
Actual Transfer	1,243	2,771	691	55
Actual Return	779	307	1,860	12
Actual Other	170	417	205	706

5. Analysis

• **Base Rate Influence:** Adoption and Transfer dominate historical intake records (>72%), providing strong prior advantages (ln P = -0.8636 and -1.2083).
• **High Precision on 'Other' (88.69%):** The 'Other' class exhibits high precision due to distinct animal intake types (e.g., bats and raccoons representing wildlife intake, which are never adopted or transferred).
• **Sex and Duration Trajectories:** Neutered/spayed status at intake strongly predicts Return to Owner (owned pets have higher sterilization rates), whereas prolonged shelter stay duration heavily correlates with eventual Adoption rather than rapid Transfer.

6. Limitations & Future Work

• **Violation of Conditional Independence:** Breed strongly implies AnimalType (e.g., Pit Bull implies Dog), causing feature redundancy that artificially inflates likelihoods.
• **Gaussian Distribution Mismatch:** Shelter stay durations follow an exponential/log-normal distribution with sharp zero-day peaks, leading to mild model misspecification under standard Gaussian assumptions.
• **Future Work:** Incorporating log-transformed durations and Tree-Augmented Naive Bayes (TAN) to capture breed-species correlations.

7. Worked Example (Test Row r = 95)

For student test index **r = 242013 % 97 = 95** (Actual Ground Truth: **Transfer**):
• **Intake Attributes:** Cat (AnimalType=0), Intact (SexUponIntake=1), Black (Color=2), Domestic Shorthair Mix (Breed=0), Age z-score = -0.7164, Stay z-score = -0.3963.
• **Categorical Likelihoods for Transfer:** ln P(Cat|Transfer) = -0.5237, ln P(Intact|Transfer) = -0.3761, ln P(Black|Transfer) = -1.3942, ln P(DSH|Transfer) = -0.7755.
• **Numeric Likelihoods:** Age Gaussian ln N = +0.1654, Stay Duration Gaussian ln N = +0.2195.
• **Log-Posteriors:** Adoption: -6.9856 (P=0.1822) | **Transfer: -5.5267 (P=0.7837)** | Return: -11.4212 (P=0.0022) | Other: -8.7274 (P=0.0319).
• **Outcome:** The model confidently and correctly predicted **Transfer** with **78.37% posterior probability**.